

Jens Laufer

Forward Deployed Engineer — LLM agents running in production, inside the systems a business already has

Python · LLM agents, tool use & MCP · agent harness (loop, memory, approval gates, observability, budget control) · RAG · REST APIs & enterprise integration (SAP, ESB, canonical data models) · Java / Spring Boot · Docker / CI

I build the harness: the loop, the tools, the memory, the gates and the observability that turn a language model into a system somebody can rely on while nobody is watching. Two of mine run in production today — an autonomous AI business partner that plans, writes code, tests and deploys around the clock, and Fabrik, a multi-agent platform where worker agents ship software in parallel. Before that, sixteen years of back-end and integration work sitting with the customer: invoice platforms, handing invoices to SAP over an ESB, master data migration at a major bank, an industry file format in utility billing. That is the forward deployed combination — the AI side, and the side where ERP, CRM and document workflows have been running for decades. Freelance since 2009, so requirements, delivery and production support have always been the same conversation.

AVAILABLE immediately	WORKS worldwide	REMOTE SHARE 95 % 5 % on-site
DAY RATE €2,000/day excl. VAT	BASED IN Karlstein am Main Germany	

FOCUS

Harness engineering, in production since 12/2025: agent loop, **tool use over MCP**, RAG, memory, approval gates, observability, budget control — the scaffolding around the model, not the model itself. Two live systems, several AI-built SaaS products shipped end to end.

Forward deployed by habit: freelance since 2009, always inside the customer's team (Telekom, Commerzbank, Volkswagen, Audi, DVAG, Brunata) — scoping in conversation, delivering on site or remote, staying through production.

Integration into systems that already exist: invoice handover to **SAP** over an ESB (1&1), master data migration on a canonical XML model (Commerzbank), the Haweiko-Arge industry format in heating cost billing (Brunata).

Prototype to production: took over a running prototype at Deutsche Telekom, developed it further and put it live — with the Docker/Kubernetes, tracing and CI work that being live actually requires.

Python and back-end depth: REST back-ends in Python (Eve, Flask) and Java/Spring Boot microservices with OpenAPI, shipped in Docker, test-driven throughout.

ROLES

- Full-stack engineering
- Solution design
- Backend engineering
- DevOps
- Build & configuration management

Data analysis

Prototyping

Machine learning

Jenkins plugin development

Maven plugin development

SKILLS

API / INTEGRATION

REST

Swagger

OpenAPI

SOAP

Webservices

CXF

XFire

Kafka

ServiceMix

DEVOPS / CLOUD

Docker

Kubernetes

Rancher

GitLab CI

Jenkins

Maven

Grunt

npm

Artifactory

AWS (EC2, S3, SageMaker, VPC)

Microsoft Azure

DigitalOcean

FRONTEND

Vue.js

React

AngularJS

GWT

Quasar

HTML

CSS

Bootstrap

SECURITY / IAM

Keycloak

Spring Security

OAuth2

MACHINE LEARNING

scikit-learn

Keras

DeepLearning4J

CNN

Transfer Learning

Supervised/Unsupervised

k-means

PCA

t-SNE

Reinforcement Learning

OpenAI Gym

Topic Modeling

TOOLING / OTHER

Git

GitLab

Subversion

Jira

Confluence

Kibana

ANTLR

JavaCC

ZXing (QR)

PayPal API

RDF/XML/JSON

SAP (BAPI)

Phonegap

JBoss

Tomcat

LANGUAGES

Java

JavaScript

TypeScript

Python

R

Groovy

MICROSERVICES / BACKEND

Spring Boot

Spring

Spring Data

Spring Integration

Spring Security

OAuth2

Microservices

Feign

Sleuth

Zipkin

Python Eve

Python Flask

Java EE

EJB

PERSISTENCE / DATABASES

MySQL

MongoDB

PostgreSQL

Oracle

Neo4j

JPA

Hibernate

TESTING

JUnit

TDD

Mockito

Selenium

DBUnit

Playwright

DATA SCIENCE

pandas

numpy

dplyr

Jupyter

RMarkdown

ggplot

seaborn

Vega-Lite

A/B Testing

Web scraping

PROJECT HISTORY

since 12/2025
ongoing

Autonomous AI agent systems — Harness & Agentic Engineering

Solytics GmbH · Remote — AI / SaaS

Harness engineering · Agent engineering · AI system architecture · Full-stack engineering · DevOps

Building and running autonomous LLM agent systems as a **harness engineer**: the scaffolding around the model — agent loop, tool use over **MCP**, approval gates, memory, observability and budget control — not the model itself. Two systems in production: a self-directing AI business partner that plans, writes code, tests and deploys around the clock, and **Fabrik** (fabrikhq.com), a multi-agent platform for autonomous software delivery with parallel worker agents. Event-driven orchestration over Telegram, GitHub and systemd; several AI-built SaaS products shipped end to end.

Claude / Anthropic API

LLM-Agenten

Agent Harness

Multi-Agent-Orchestration

Tool Use / MCP

RAG

Prompt Engineering

Python

FastAPI

Vue 3

Go

Docker

systemd

GitHub Actions

DigitalOcean

SQLite

TDD

TypeScript

07/2024–06/2025
12 months

Digital property profile for owners

verkaufe.immobilien · Remote — Internet

Full-stack engineering · Solution design · DevOps

Full-stack development of a digital property profile that owners create and maintain themselves: Vue.js front-end, Python back-end (Eve) with a REST API and MongoDB. Day-to-day GitLab work (version control, merge-request and review workflow); built and maintained the GitLab CI pipelines for automated build and deployment. Worked to DevOps principles, owning build and operation of the services in Docker. Test-driven development, automated end-to-end tests with Playwright.

Python

Python Eve

MongoDB

TDD

Vue.js

Docker

GitLab

GitLab CI

REST

Playwright

07/2022–04/2024
22 months

Front- and back-end, heating cost billing

Brunata · Remote — Industry

Full-stack engineering · Solution design · DevOps

Production server-side services and APIs in Java and Spring Boot: a microservice that parses and generates heating-cost files in the Haweiko-Arge industry format. Designed and built in a microservices architecture with REST interfaces (OpenAPI, Swagger), secured with Spring Security and Keycloak, persisted in MongoDB. Day-to-day GitLab work (version control, merge-request and review workflow); built and maintained the GitLab CI pipelines for automated build and deployment. Worked to DevOps principles, owning build and operation of the services in Docker. Test-driven development (JUnit, Mockito), automated end-to-end tests with Playwright. Front-end development.

Java

Spring Boot

MongoDB

JUnit

TDD

React

Docker

GitLab

GitLab CI

Spring Security

Keycloak

Mockito

OpenAPI

Swagger

REST

Microservices

Playwright

01/2021–03/2021

3 months

Marketing campaign tooling

DigitalMarketer.com · Remote — Marketing

Full-stack engineering · Solution design

Tools that generate marketing campaigns and publish them to Google Ads and Twitter Ads: Java/Spring Boot microservices with REST interfaces (OpenAPI, Feign), Vue.js/Quasar front-end, storage in AWS S3. Designed and built in a microservices architecture. Day-to-day GitLab work (version control, merge-request and review workflow); built and maintained the GitLab CI pipelines. Worked to DevOps principles, owning build and operation of the services in Docker and Kubernetes. Test-driven development (JUnit, Mockito).

Scrum Spring Boot Docker Kubernetes Microservices REST Java OpenAPI Feign JUnit TDD
Mockito Vue.js Quasar AWS S3 GitLab CI GitLab

12/2019–12/2020

13 months

Web-based configuration of telecommunications devices

Deutsche Telekom · Darmstadt/Remote — Telecommunications

DevOps · Build & configuration management · Full-stack engineering

Took over an existing prototype for configuring telecommunications devices over a web interface, developed it further and moved it into production. Production server-side services and APIs in Java and Spring Boot in a microservices architecture (REST, Feign, Spring Data/JPA), Vue.js front-end, secured with Keycloak, persisted in MySQL and MongoDB. Worked to DevOps principles, owning build, operation and monitoring of the services: Docker and Kubernetes (Rancher), logging and tracing via Kibana, Sleuth and Zipkin. Day-to-day GitLab work (version control, merge-request and review workflow); built and maintained the GitLab CI pipelines for automated build and deployment. Test-driven development (JUnit, Mockito).

Java Spring Boot Docker Kubernetes Microservices REST Spring Data JPA JUnit TDD
Mockito Vue.js MySQL MongoDB HTML CSS GitLab GitLab CI Rancher Kibana Feign
Keycloak Sleuth Zipkin

07/2015–07/2017

25 months

Invoice management platform (SME)

Deutsche Verrechnungstelle / DVAG · Frankfurt a. Main — Insurance, Finance, Startup

Full-stack engineering

Platform covering the full invoice life cycle (creation, payment, dunning, debt collection); a range of Spring Boot microservices.

Spring Boot Docker Kubernetes Microservices REST Java Maven Spring Security OAuth2
Spring Integration MongoDB Jira Confluence JUnit TDD Mockito Selenium AngularJS Grunt
npm Spring Git

07/2019–12/2019

6 months

Training a stock-trading agent

JiGaoTrading · Remote — Finance

Machine learning · Data analysis

An agent that trades autonomously on Chinese stock exchanges (reinforcement learning).

Reinforcement Learning Python R Docker AWS S3 OpenAI Gym GitLab seaborn AWS SageMaker
MongoDB

04/2010–09/2010
6 months

Web services on an ESB (mediation and rating)

1&1 · Montabaur — Telecommunications
Backend engineering

Software design, web services, maintenance of the rating engine; web services that hand invoices over to **SAP**.

- Java
- Spring
- Spring Security
- JPA
- Hibernate
- Maven
- JUnit
- TDD
- Mockito
- DBUnit
- ServiceMix
- Webservices
- CXF
- XFire
- SAP-Anbindung
- SAP BAPI
- JBoss
- Oracle
- Subversion (SVN)

05/2009–03/2010
11 months

Master data migration at a major bank

Commerzbank · Frankfurt a. Main — Banking, Finance
Backend engineering

Workflows for persisting master data, a canonical XML model, RDF description for machine processing.

- Java
- Java EE
- IBM MDM (WCC)
- EJB
- JPA
- RDF
- XML
- Oracle
- Subversion (SVN)

CONTACT & CREDENTIALS

CONTACT & PROFILES

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PUBLICATION	Towards Data Science — author

EDUCATION

11/1997	Dipl.-Ing. (Univ.) Electrical Engineering (MSc equivalent) University of Erlangen-Nuremberg
05/1989	Abitur (university entrance qualification) Gymnasium am Romäusring, Villingen-Schwenningen

CERTIFICATES

12/2018	Machine Learning Engineer Nanodegree Udacity Verify →
08/2017	Data Analyst Nanodegree Udacity Verify →